

BPM1 FTM IMCQ 8

Fall 2025 - KEY



1. A 22-year-old man is brought to the emergency department after sustaining a traumatic injury during a football match. He reports hearing a sudden "pop" in his right knee, followed by immediate sharp, localized pain and inability to bear weight. His temperature is 36.8°C (98.2°F), pulse is 88/min, respirations are 16/min, and blood pressure is 122/74 mm Hg. Physical examination shows significant swelling and bruising over the right knee. There is a marked tenderness to palpation of the lateral joint line. The patient localizes the pain precisely to the lateral aspect of the knee, describing it as “stabbing” and exacerbated by movement. MRI confirms a complete tear of the anterior cruciate ligament (ACL) and lateral meniscus. Which of the following nerve fibers is most likely responsible for transmitting the pain?

- A. Somatic afferents
- B. Visceral afferents
- C. Sympathetic efferents
- D. Parasympathetic efferents
- E. Visceral efferents

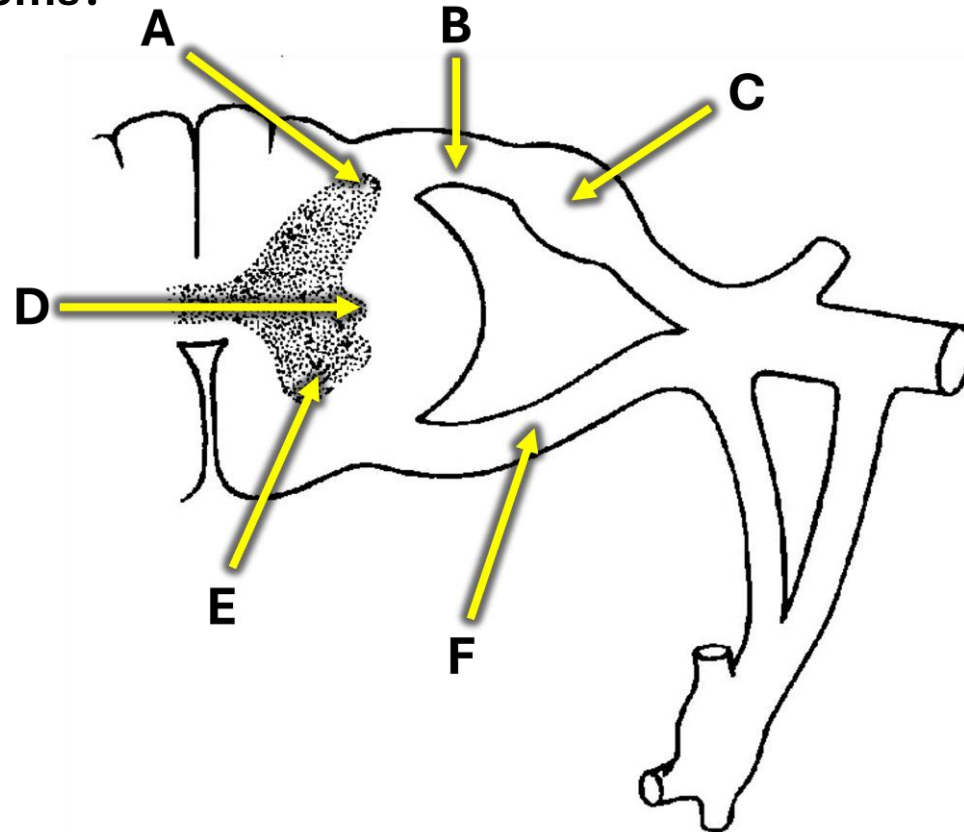
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SOM.MK.II.BPM1.1.FTM.1.ANAT.1012 Describe the basic organization and function of the spinal cord and spinal nerves

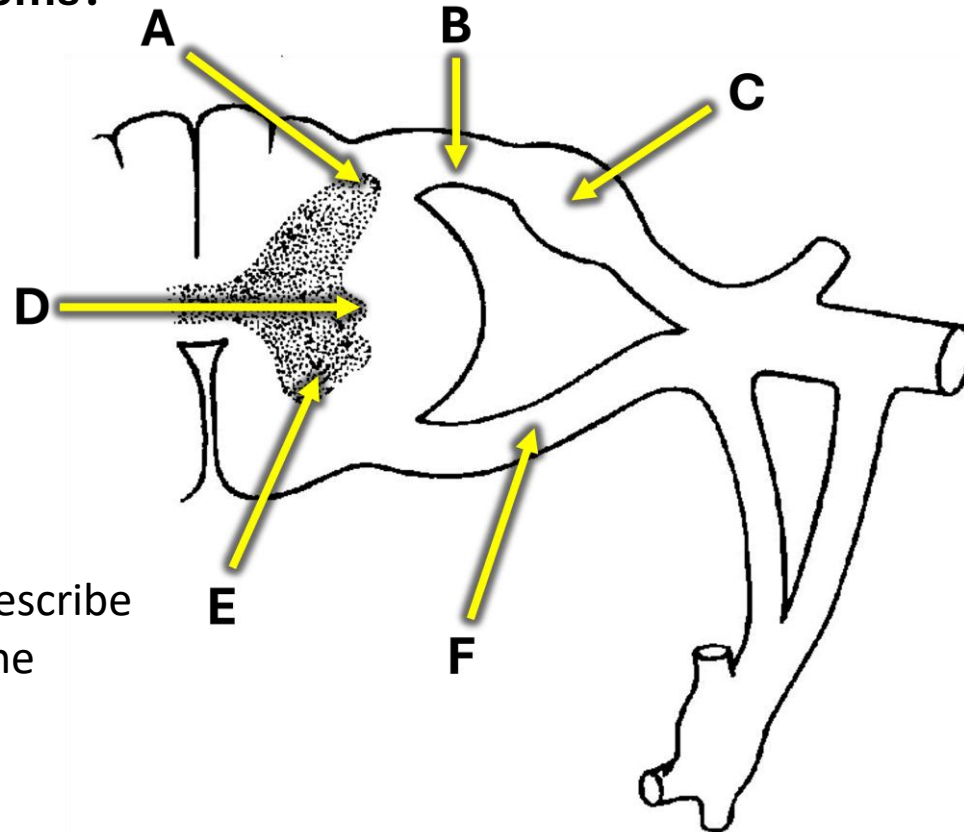
2. A 62-year-old man comes to the physician because of a 9-month history of progressive muscle weakness. He reports frequent muscle twitching. His wife notes he has become increasingly clumsy and struggles to button his shirts. Physical examination shows diffuse muscle atrophy and hyperreflexia in all limbs. MRI of the brain and spine shows no structural lesions. Electromyography (EMG) shows widespread denervation suggestive of motor neuron disease. Which of the following labeled regions most likely contain the cell bodies of the neurons responsible for his symptoms?

- A. A
- B. B
- C. C
- D. D
- E. E
- F. F



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SOM.MK.II.BPM1.1.FTM.1.ANAT.1012 Describe the basic organization and function of the spinal cord and spinal nerves

3. A 40-year-old man is brought to the emergency department by his wife because of a 3-hour history of nausea, vomiting, watery diarrhea, profuse sweating, blurred vision, and “watery eyes”. His wife notes he was spraying pesticides on their farm earlier without protective gear. His temperature is 36.7°C (98.1°F), pulse is 52/min, respirations are 18/min, and blood pressure is 90/60 mmHg. He appears diaphoretic and anxious. Physical examination shows pinpoint pupils, excessive lacrimation and salivation, diffuse wheezing, and hyperactive bowel sounds. Based on the physical examination findings for the pulse and abdominal sounds, which of the following is most likely mediating its parasympathetic activity?

- A. Olfactory nerve (CN I)
- B. Optic nerve (CN II)
- C. Trigeminal nerve (CN V)
- D. Vestibulocochlear nerve (CN VIII)
- E. Vagus nerve (CN X)
- F. Hypoglossal nerve (CN XII)

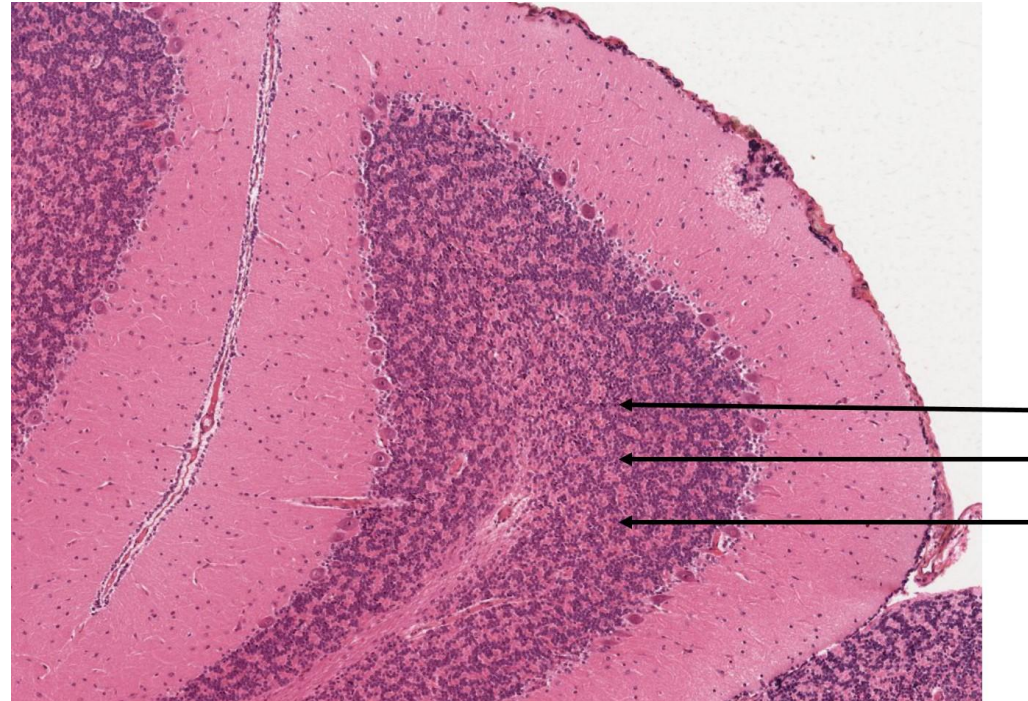
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SOM.MK.II.BPM1.1.FTM.1.ANAT.1015 List the four cranial nerves that contain preganglionic parasympathetic fibers.

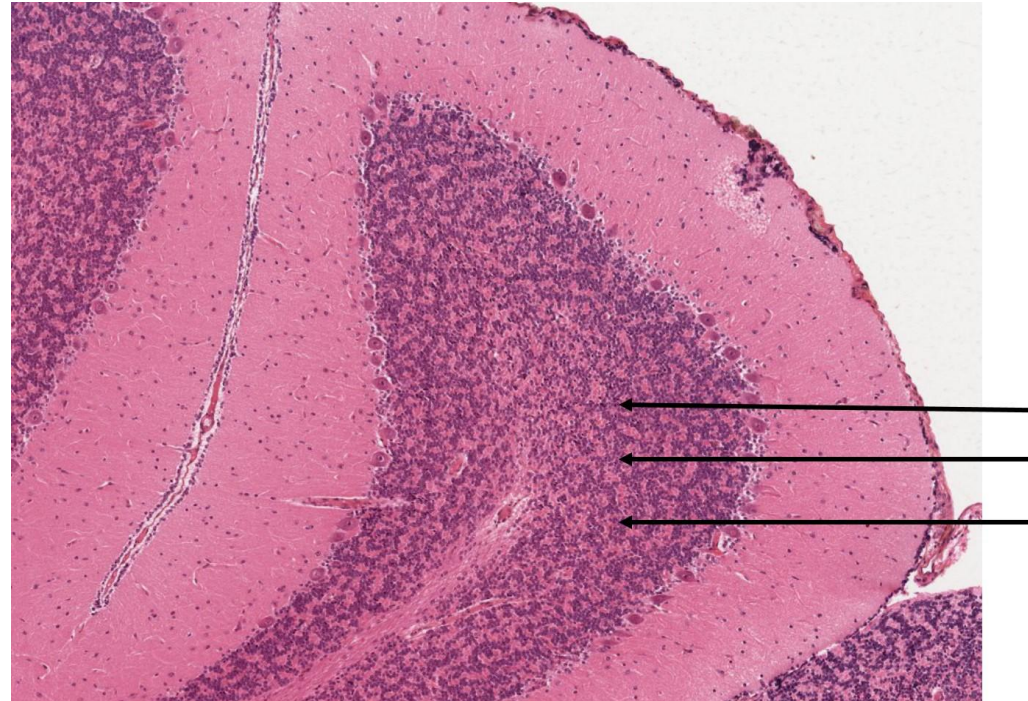
4. A 65-year-old man is brought to the emergency department because of a 2-week history of an unsteady gait and loss of balance when standing. He has a history of alcoholism. Physical examination shows a wide gait, inability to stand with his eyes closed, dysmetria and slurred speech. An MRI of his brain shows a tumor in the region shown at the tips of the arrows in the attached photomicrograph. Which of the following cells best characterizes this region?

- A. Pyramidal
- B. Purkinje
- C. Granule
- D. Oligodendrocytes
- E. Basket



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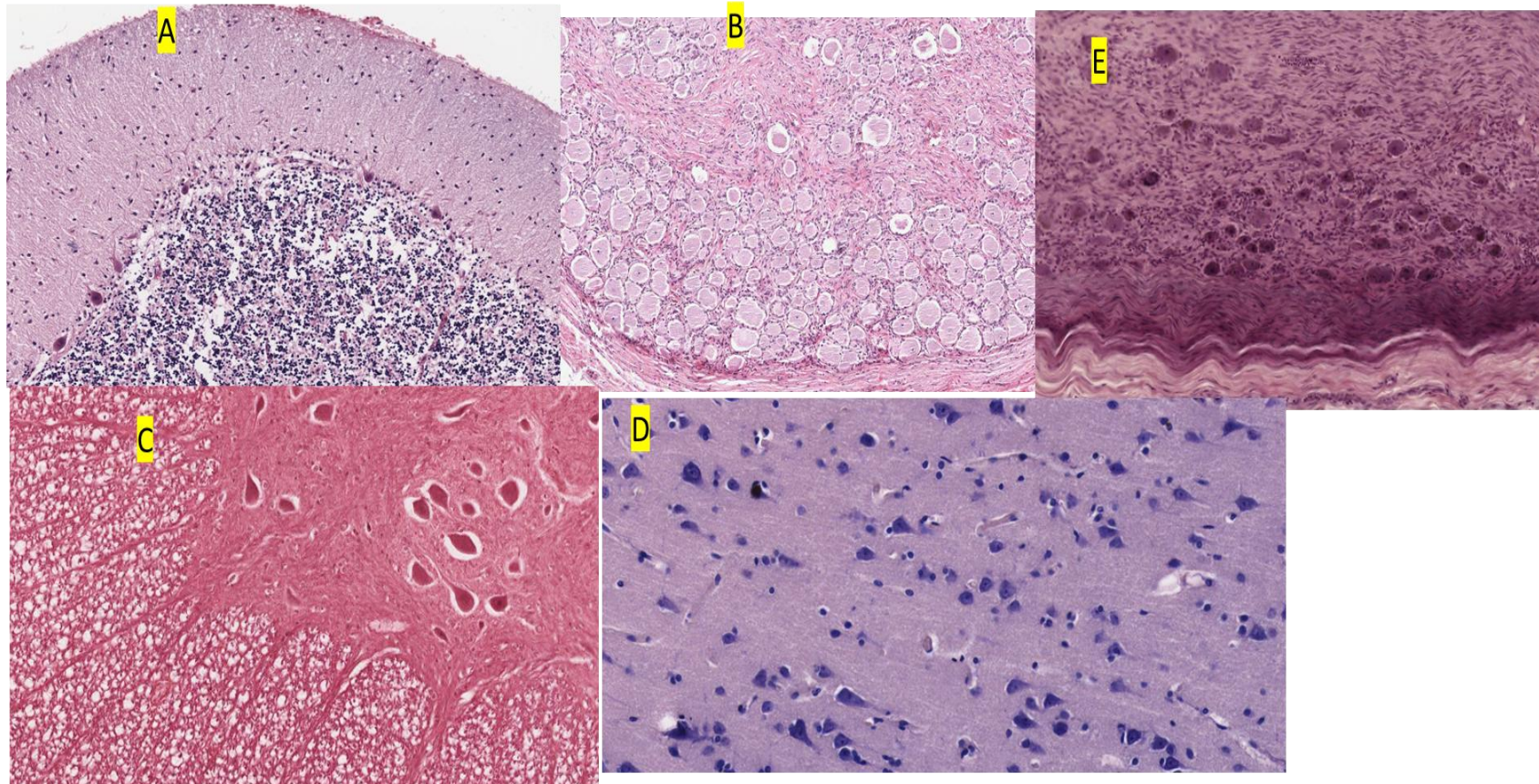
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Describe the microscopic structure of the cerebellar cortex.
SOM.MK.I.BPM1.1.FTM.3.HCB.0966

5. A 65-year-old man is brought to the emergency department because of a 1-week history of slurred speech and difficulty moving his left arm and leg. Physical examination shows weakness in the left extremities. Imaging studies shows an occlusion of a major vessel supplying the cerebrum. Which of the following images best corresponds to the region most likely to be affected by this occlusion?

- A. A
- B. B
- C. C
- D. D
- E. E



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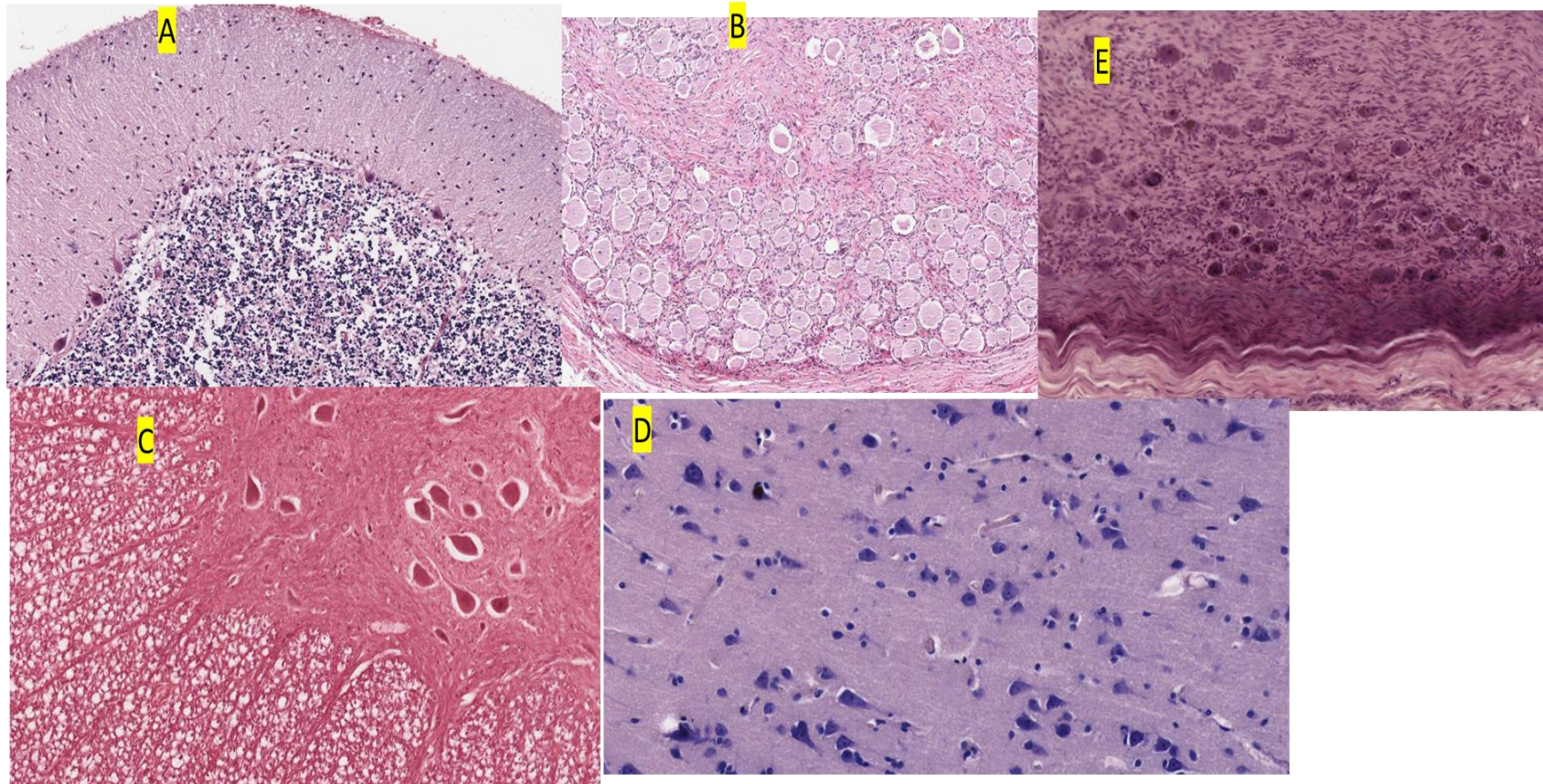
A. A

B. B

C. C

D. **D**

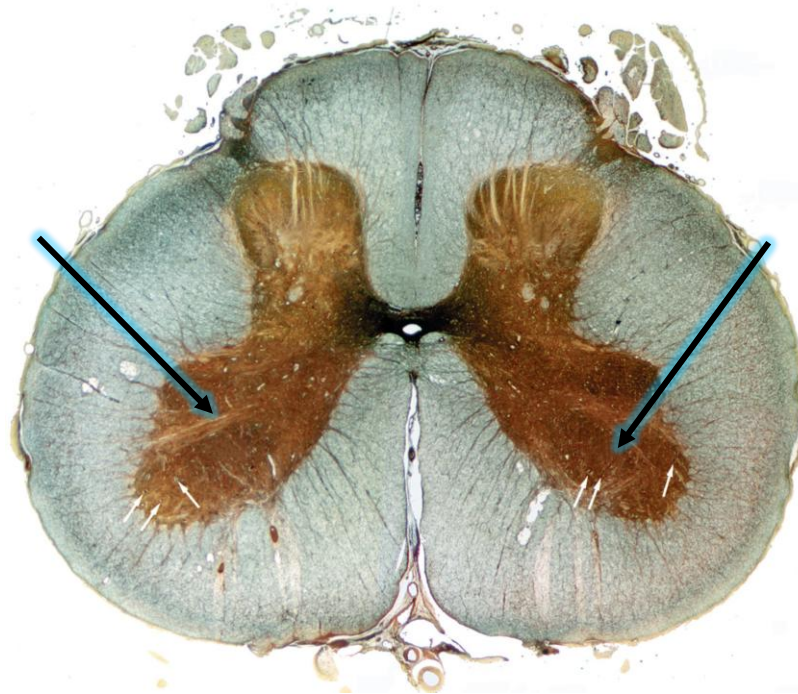
E. E



Describe the microscopic structure of the cerebral cortex, including characteristics of the pyramidal cells. SOM.MK.I.BPM1.1.FTM.3.HCB.0964

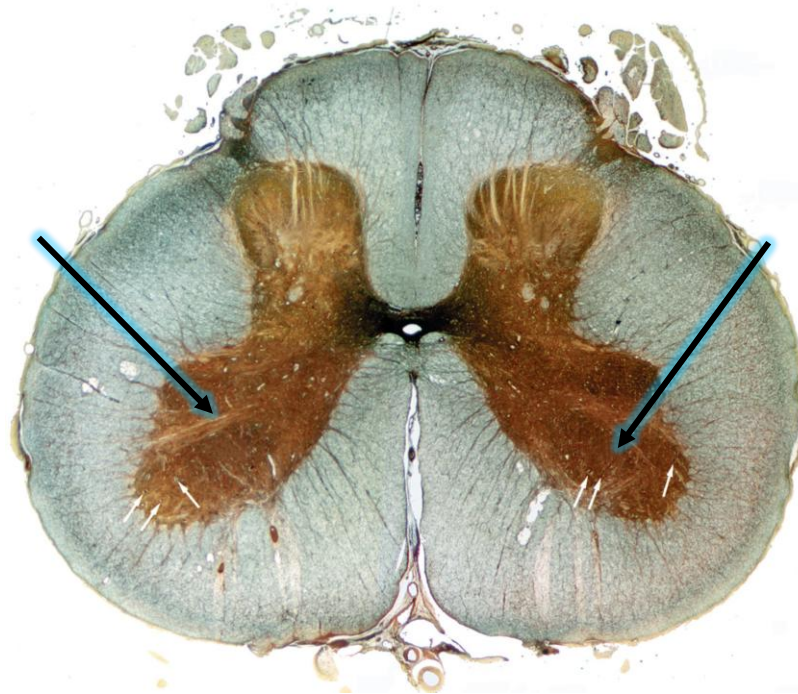
6. A 34-year-old man is brought to the emergency department because of a 1-hour history of inability to move his limbs. He was involved in a car crash a few hours ago. Physical examination shows loss of muscle power in all his limbs. MRI shows trauma to the regions at the pointer in the attached photomicrograph. Which of the following best characterizes the cells most likely to be affected?

- A. Multiple processes from the cell body
- B. Carries somatic afferent fibers
- C. A single process extending from the cell body
- D. Forms a site for visceral efferent neurons
- E. Two processes from the cell body



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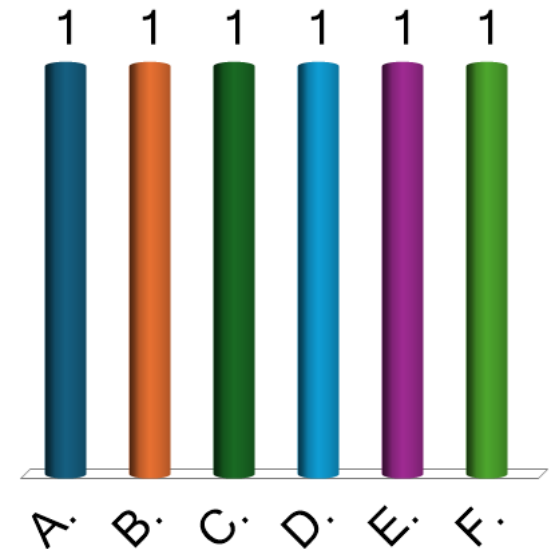
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Describe the histological appearance of the gray matter of the spinal cord, including characteristics of alpha motor neurons. SOM.MK.I.BPM1.1.FTM.3.HCB.0970

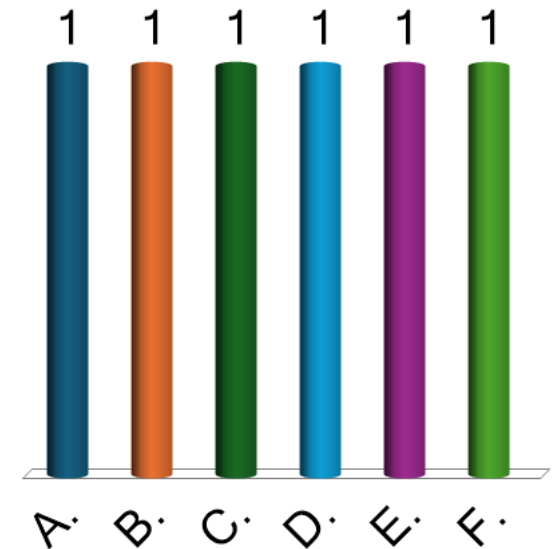
7. A 26-year-old woman is brought to the emergency department 45 minutes after she was involved in a motor vehicle collision. She is bleeding from multiple wounds on her extremities. Her pulse is 122/min and blood pressure is 96/64 mm Hg. Physical examination shows altered mental state, multiple fractures of limbs, skin lacerations, cold extremities, dilated pupils, and sweating in the palms, soles, and face. She is resuscitated with intravenous fluids and provided with supportive treatment. Increased activity of which of the following autonomic nerves to the iris best explains dilated pupils in this patient?

- A. Preganglionic sympathetic adrenergic nerves
- B. Postganglionic sympathetic adrenergic nerves
- C. Postganglionic parasympathetic cholinergic nerves
- D. Postganglionic sympathetic cholinergic nerves
- E. Preganglionic parasympathetic cholinergic nerves
- F. Preganglionic parasympathetic adrenergic nerves



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- F. Preganglionic parasympathetic adrenergic nerves



8. A 23-year-old woman comes to the physician because of a 6-month history of recurrent episodes of palpitations, chest tightness, nervousness, difficulty breathing, and sweating. She says each episode peaks in about 10 minutes and lasts approximately 30-40 minutes and resolves spontaneously without any medications. Her vital signs are within normal limits. Physical examination shows no abnormalities. Results of laboratory studies are within the reference ranges. She is diagnosed with panic disorder. Activation of the sympathetic nervous system is responsible for her symptoms during an episode. Activation of which of the following receptors best explains sweating during an episode in this patient?

- A. Alpha1
- B. Alpha2
- C. Beta1
- D. Beta2
- E. M2
- F. M3

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- D. Beta2
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- ✓ F. M3

SOM.MK.II.BPM1.1.FTM.3.PHYS.0044 Describe the location and function of cholinergic and adrenergic receptor subtypes and the signaling mechanisms associated with them.

9. A 6-year-old girl is brought to the emergency department by her mother because of a 15-minute history of shortness of breath and wheeze. The mother says she had a bee sting while playing in the yard, 30 minutes ago. She has allergic dermatitis since birth. Her pulse is 132/min, blood pressure is 90/56 mm Hg, and respirations are 36/min. Physical examination shows erythematous skin rash on the face, trunk, and extremities. Pulmonary examination shows rhonchi (gurgling breath sounds) in all lung fields bilaterally. She is diagnosed with anaphylaxis. She receives an intramuscular injection of epinephrine and 30% oxygen inhalation through a venturi mask. This drug activates beta adrenergic receptors in the bronchial smooth muscles. Her symptoms improve. Which of the following signaling mechanisms best explains the relaxation of bronchial smooth muscles in this patient?

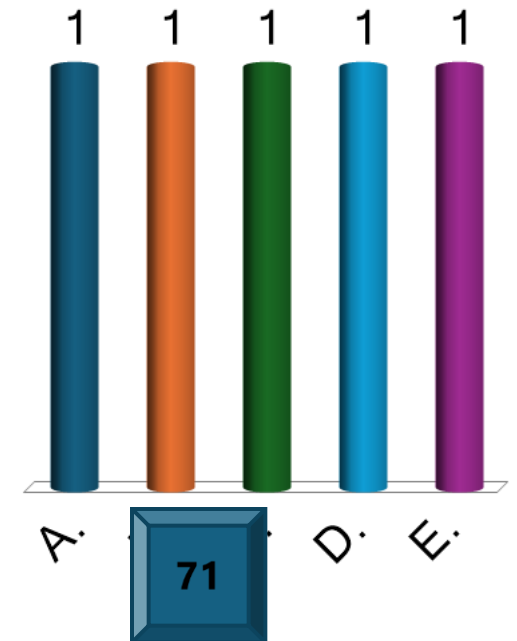
- A. Gs protein mediated increased cGMP
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- C. Gs protein mediated increased cAMP
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- E. Gi protein mediated decreased cAMP

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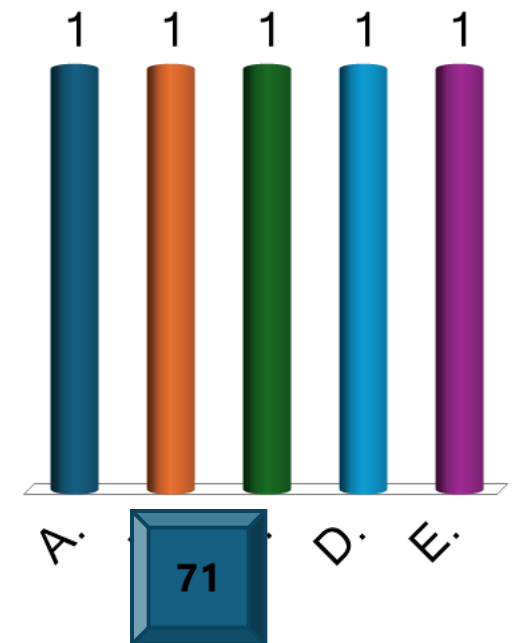
10. A 43-year-old man is brought to the emergency department by his wife because of a 1-hour history of nausea, blurred vision, increased tears, muscle weakness, and noisy breathing (wheeze). The wife says that he was spraying insecticides in the garden, 2 hours ago. His temperature is 36.4°C (97.5°F) and blood pressure is 110/62 mm Hg. Physical examination shows confusion, moist skin with excessive sweating, and drooling. Pulmonary examination shows a high-pitched sound during expiration in all lung fields. He is diagnosed with organophosphate poisoning and provided with supportive treatment. Which of the following additional physical findings is most likely in this patient?

- A. Dilated pupils
- B. Piloerection
- C. Increased pulse rate
- D. Dry skin
- E. Constricted pupils



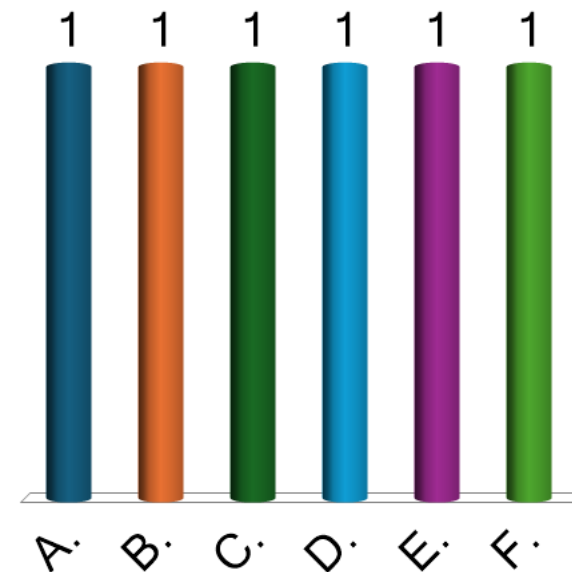
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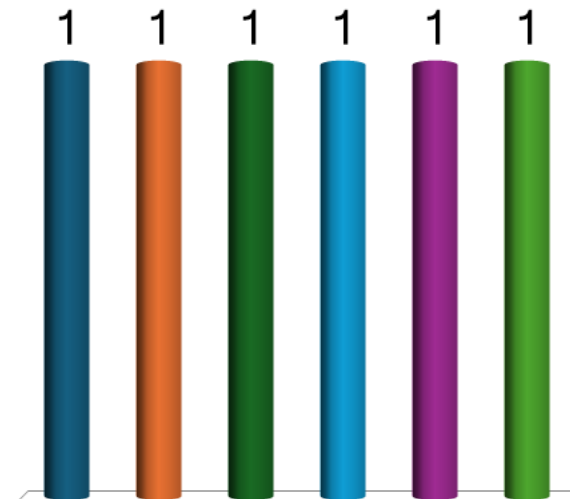
11. A 59-year-old man comes to the physician because of a 4-month history of dry cough and 4-kg (8.8-lb) weight loss. He has smoked 1 pack of cigarettes daily for 30 years. Vital signs are within normal limits. Physical examination shows a constricted right pupil, ptosis of the right upper eyelid, and reduced sweating on the right side of the face. A CT scan of the chest shows a mass in the right apical region. He is diagnosed with a lung tumor that is compressing and damaging several structures, including the superior cervical sympathetic ganglion on the right side. Which of the following structures is most likely innervated by the postganglionic fibers arising from this ganglion in this patient?

- A. Constrictor pupillae muscle of the iris
- B. Ciliary muscle of the right eye
- C. Dilator pupillae muscle of the iris
- D. Heart muscles
- E. Bronchial smooth muscles
- F. Urinary bladder

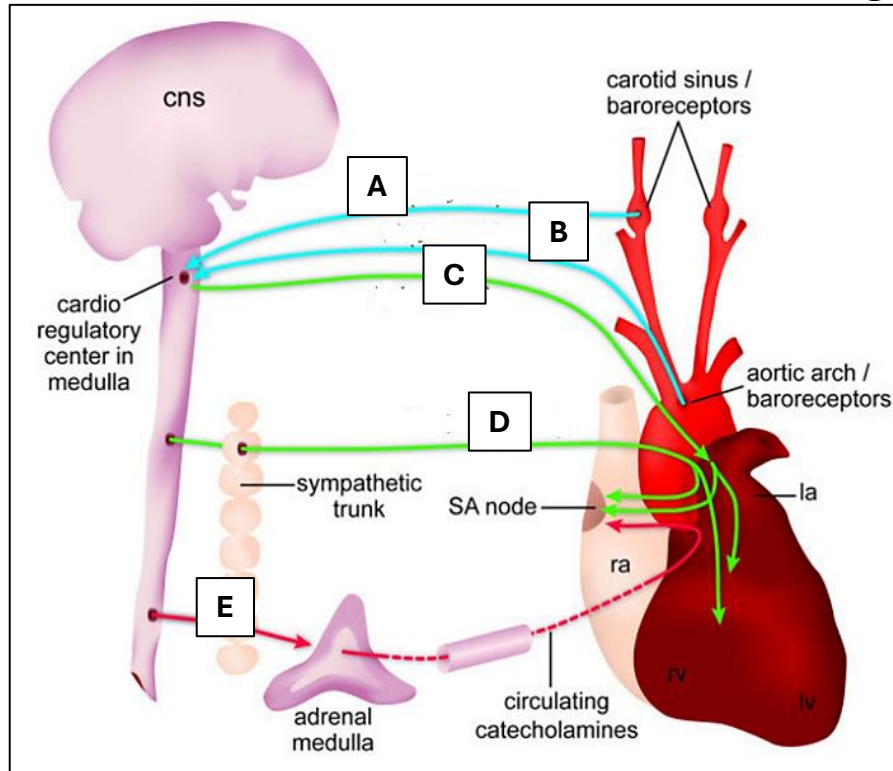


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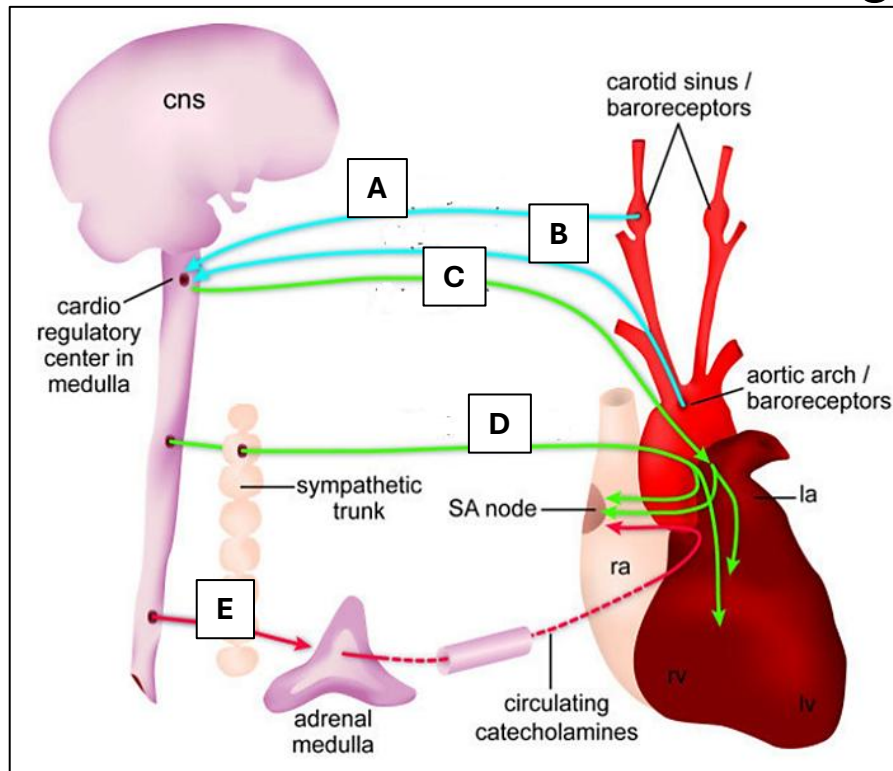


12. A 55-year-old man is brought to the emergency department 30 minutes after sustaining multiple gunshot wounds to his limbs. He is bleeding and found to have lost a large volume of blood. His pulse is 124/min, blood pressure is 96/60 mm Hg, and respirations are 22/min. He is diagnosed with hypovolemic shock due to blood loss. The baroreflex mechanism of blood pressure regulation is active in this patient with generalized stimulation of the sympathetic nervous system. Which of the following neurotransmitters is most likely released from the post-ganglionic nerve labelled, **D** in the figure in this patient?



- | | |
|-------------------|---|
| A. Epinephrine | 1 |
| B. Norepinephrine | 1 |
| C. Dopamine | 1 |
| D. Acetylcholine | 1 |
| E. Nitric oxide | 1 |

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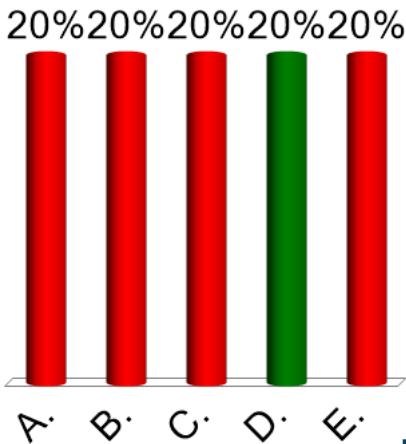
SOM.MK.II.BPM1.1.FTM.3.PHYS.0048 List the functions, afferent and efferent pathways and stimuli of the baroreflex. Predict clinical implications of baroreflex dysfunction.

13. A 25-year-old man participates in a study related to the neurotransmitter released and the postsynaptic receptor involved in the autonomic nervous system. His pulse is 78/min and blood pressure is 124/82 mm Hg. The right vagus nerve (cranial nerve X) is identified and stimulated for a few seconds. Electrical stimulation of this nerve activates preganglionic autonomic nerves passing through this nerve and the changes observed are shown:

Pulse is 56/min, blood pressure is 110/70 mm Hg, bronchoconstriction, and increased gut motility.

Which of the following combinations best explains reduction in pulse rate following nerve stimulation in this volunteer?

Option	Neurotransmitter Released	Postganglionic Receptor type	Second Messenger involved
A	Acetylcholine	M3	cAMP
B	Norepinephrine	Alpha1	IP3
C	Norepinephrine	Beta1	cAMP
D	Acetylcholine	M2	cAMP
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SOM.MK.II.BPM1.1.FTM.3.PHYS.0044 Describe the location and function of cholinergic and adrenergic receptor subtypes and the signaling mechanisms associated with them.

14. A 54-year-old man is brought to the emergency department because of a 2-history of hallucinations, confusion, vomiting and difficulty breathing. His symptoms started during the intermission of an opera play. Several other patients were admitted to nearby emergency rooms with similar symptoms. The patient has a 7-year history of hypertension and hyperlipidemia treated with the appropriate pharmacotherapy. His temperature is 37°C (98.6°F), blood pressure is 108/68 mm Hg, pulse is 32/min, and respirations are 28/min. Physical examination shows severely decreased muscle strength in all extremities, pinpoint pupils, increased salivation and lacrimation and wheezing in all lung fields bilaterally. The drug of choice is administered for his condition, and his symptoms improve except the muscle weakness. Which of the following most likely explains the persistent muscle weakness after the administration of the drug of choice?

- A. Muscarinic receptor overstimulation
- B. Adrenergic beta-1 receptor blockade
- C. Nicotinic receptor muscle type overstimulation
- D. Muscarinic receptor blockade
- E. Increased activity of acetylcholinesterase

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SOM.MK.1.BPM1.1.FTM.4.PHAR.0044

Describe the organ system effects, mechanism of action, uses and adverse effects of indirect-acting cholinergic agonists (anticholinesterases). Compare and contrast the effects of direct-acting and indirect-acting cholinergic agonists on the cardiovascular system.

SOM.MK.1.BPM1.1.FTM.4.PHAR.0046: Describe the organ system effects, mechanism of action, therapeutic uses, adverse effects and contraindications of muscarinic receptor antagonists.

15. A team of pharmacologists is studying the effects of hexamethonium on different tissues in a model animal. A bolus of the drug is administered. Which of the following sets of physiological changes are most likely to be observed at the effector sites below?

- A. Miosis, bradycardia, anhidrosis
- B. Vasoconstriction, hypersalivation, mydriasis
- C. Urinary retention, miosis, diarrhea
- D. Tachycardia, cycloplegia, xerostomia
- E. Venodilation, tachycardia, urinary incontinence

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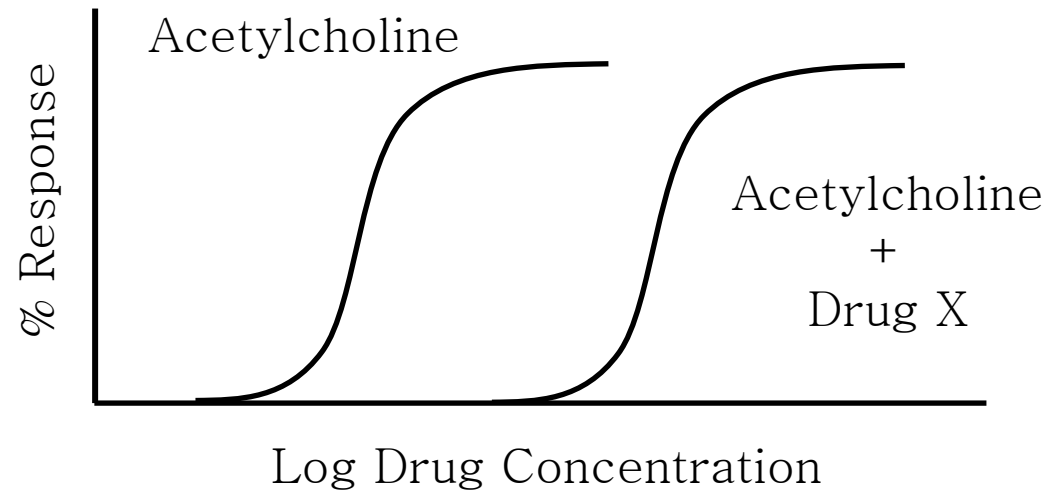
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- E. Venodilation, tachycardia, urinary incontinence

SOM.MK.1.BPM1.1.FTM.4.PHAR.0047 Describe the mechanism of action of the ganglion blockers and predict their effects by applying knowledge of the predominant tone of a dually-innervated organ.

Nicotinic Receptor Antagonists

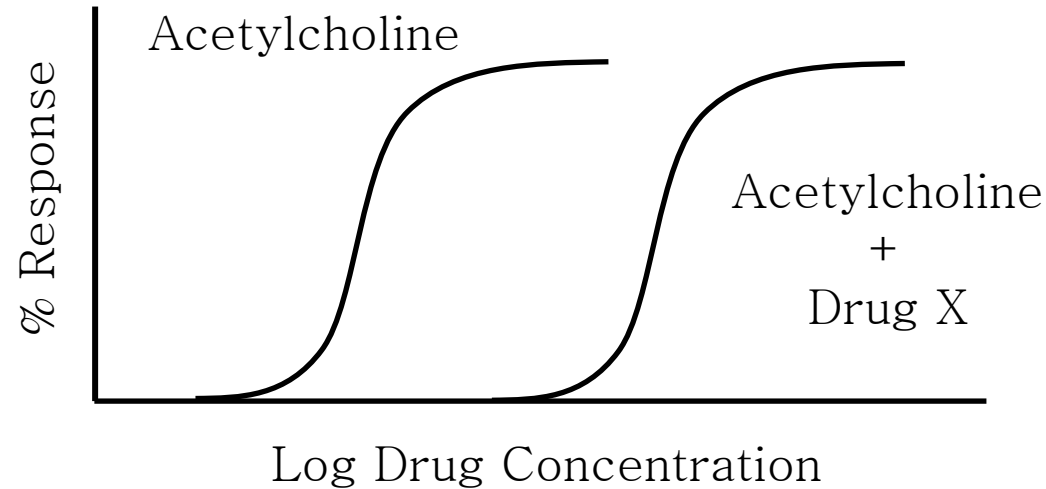
Effector Site	Predominant Tone	Receptor Type	Predominant Tone Effect	Effect of Ganglionic Blockade
Arterioles	Sympathetic (adrenergic)	$\alpha 1$	Vasoconstriction; hypertension	Vasodilation; Hypotension
Veins	Sympathetic (adrenergic)	$\alpha 1$	Venoconstriction	Venodilation
Heart	Parasympathetic (cholinergic)	M2	↓Heart Rate	↑Heart Rate
Iris	Parasympathetic (cholinergic)	M3	Miosis	Mydriasis
Ciliary muscle	Parasympathetic (cholinergic)	M3	Accommodation to near vision	Cycloplegia (focus for far vision)
GI tract	Parasympathetic (cholinergic)	M3	↑tone and motility; diarrhea; ↑secretions	↓ tone and motility; constipation; ↓ Secretions
Urinary bladder	Parasympathetic (cholinergic)	M3	Urinary incontinence	Urinary retention
Salivary glands	Parasympathetic (cholinergic)	M3	Hypersalivation	Xerostomia
Sweat glands	Sympathetic (cholinergic)	M3	Hyperhidrosis	Anhidrosis

16. A team of pharmacologists is studying the effects of a novel drug, drug X, on muscarinic receptors. The concentration response curves of acetylcholine and drug X plus acetylcholine on muscarinic receptors are shown in the diagram below. Which of the following drugs is most likely similar with drug X?



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- B. Atropine
- C. Physostigmine
- D. Hexamethonium
- E. Pilocarpine

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SOM.MK.1.BPM1.1.FTM.4.PHAR.0046: Describe the organ system effects, mechanism of action, therapeutic uses, adverse effects and contraindications of muscarinic receptor antagonists.